

Thandolwethu Shakes Dlamini

RENEWABLE ENERGY FINANCE & DATA INFRASTRUCTURE | GEOSPATIAL ANALYTICS | AI ENGINEERING

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Professional Summary

Built geospatial analytics and data infrastructure for a 200K+ solar asset portfolio: 7 automated ETL pipelines integrating operational, financial, and geographic datasets with 99.9% reliability. MIT systems engineer (computational modeling) and Duke mechanical engineer with strong Python, SQL, geospatial analytics, and statistical modeling foundations.

Work Experience

Senior Data Analyst - Revenue Management | SunStrong Management LLC | Remote Aug 2025 - Present

- Led data systems migration through corporate acquisition with zero disruption to analytics across 200K+ solar assets
- Delivered quarterly performance reports and compliance filings for investors managing \$1B+ in renewable energy assets
- Maintained 97.3% data accuracy across multi-state portfolio analytics by adapting Energy Community system to IRS Notice 2025-31

Senior Data Analyst - Finance | Sunnova Energy | Remote Jun 2024 - Jul 2025

- Architected geospatial analytics platform (Python, GeoPandas) processing 200K+ assets across 29 states, unlocking \$250M+ in ITC credits by identifying 70,000 qualifying systems (\$2.5B value)
- Built 7 automated ETL pipelines (Python, Dagster, DuckDB) integrating operational, financial, and geographic datasets into unified analytics layer with 99.9% reliability
- Delivered portfolio-wide performance metrics and operational KPIs through automated reporting across Finance

Data Analyst - Financial Operations | Sunnova Energy | Remote Nov 2022 - Jun 2024

- Recovered \$95M in revenue through automated data pipelines (Python, SQL, Tableau) reconciling discrepancies across 4 data sources on unified PostgreSQL architecture
- Designed SQL/Tableau pipeline identifying 3,000 underperforming systems from 200K+ assets, delivering weekly executive dashboards to CFO

Education

Massachusetts Institute of Technology | S.M. in Technology and Policy (Systems Engineering) Sep 2020 - Sep 2022

- Computational modeling of atmospheric mercury transport and deposition

Duke University | B.S. in Mechanical Engineering & Minor in German Aug 2015 - May 2020

- Departmental Distinction | Mastercard Foundation Scholar | Pratt Fellow | Pi Tau Sigma | Allan Gray Fellow | UWC Davis Scholar

Selected Projects

AI Systematic Review — LangGraph multi-agent pipeline (GPT-4.1 + Claude), 99.3% recall across 6,673 papers at \$1.16/review. Co-authored with Johns Hopkins Dept. of Plastic and Reconstructive Surgery, in peer review at PRS.

Qualitative NLP — Privacy-preserving pipeline for on-device open-source LLM qualitative analysis of patient transcripts.

wipnote — Local-first observability and coordination platform for AI coding agents. HTML files as structured graph storage, SQLite cache. Go binary, open source.

Mercury Transport Modeling — Computational simulation of atmospheric mercury variability and regional inversion modeling. Two papers: one resubmitted to JGR: Atmospheres, one in peer review. International collaboration (MIT, CSIRO, TU Munich).

Skills

Analytics & Geospatial

Python (Advanced), SQL (Advanced), **Statistical Analysis**, **GeoPandas**, ArcGIS, LangGraph, AI Agent Systems

Engineering & Cloud

ETL Pipelines, **Dagster**, **DuckDB**, Snowflake, PostgreSQL, Tableau, Docker, AWS, GCP